

Networking Basics

1. OSI 7 Layers and Common Protocols

Layer #	Layer Name	PDU	Defender's Function & Key Protocols
7	Application	Data	User and app interface. Protocols: HTTP, HTTPS, DNS, SSH, SMB, FTP, SMTP, RDP
6	Presentation	Data	Data formatting and encryption. Formats: TLS/SSL, ASCII, JSON, MIME
5	Session	Data	Session setup and persistence. Protocols: NetBIOS, RPC, PPTP
4	Transport	Segment / Datagram	Host-to-host delivery. Protocols: TCP (reliable, 3-way handshake), UDP (connectionless)
3	Network	Packet	Logical IP routing. Protocols: IPv4, IPv6, ICMP, IPsec, BGP
2	Data Link	Frame	Physical MAC addressing and switching. Protocols: Ethernet, Wi-Fi, ARP, VLAN
1	Physical	Bit	Physical bit transmission via cables, fiber, radio, NICs, and hubs.

2. 10 Common Ports Table

Port	Protocol	Transport	Service & Defender Perspective
21	FTP	TCP	File Transfer Protocol. Cleartext transfers; check for unauthorized exfiltration.
22	SSH	TCP	Secure Shell. Encrypted remote CLI; check for brute force and lateral movement.
23	Telnet	TCP	Unencrypted text communications; legacy and insecure protocol.
25	SMTP	TCP	Simple Mail Transfer Protocol. Email delivery; check for spam bots.
53	DNS	UDP / TCP	Domain Name System. Name resolution; monitor for tunneling and C2 beacons.
80	HTTP	TCP	Hypertext Transfer Protocol. Cleartext web; inspect for exploits and downloads.
443	HTTPS	TCP	HTTP over TLS/SSL. Encrypted web; analyze certificate and traffic patterns.
445	SMB	TCP	Server Message Block. File sharing; common vector for ransomware and lateral movement.
3389	RDP	TCP / UDP	Remote Desktop Protocol. GUI administration; target for unauthorized external access.
8080	HTTP-Alt	TCP	Alternative Web Port. Used for web proxies, testing, and custom services.

3. DNS Resolution Flow

1. Local Cache: Client checks local resolver cache and hosts file.
2. Recursive Resolver: If not cached, query goes to the configured DNS resolver (e.g., 8.8.8.8).
3. Root Nameserver (.): Directs resolver to the corresponding TLD server.
4. TLD Nameserver (.com): Directs resolver to the authoritative nameserver for the domain.
5. Authoritative Nameserver: Returns the actual IP address (A/AAAA record).

6. Response & Caching: Recursive resolver caches the result and returns IP to client.

4. Normal vs. Suspicious Traffic (Port 53 / DNS)

- Normal Traffic: Standard, periodic UDP queries to designated internal/trusted DNS servers with short domain names and small payload size.
- Suspicious Traffic: High-frequency queries sent directly to an external unauthorized IP, containing long encoded subdomains (DNS tunneling for data exfiltration or C2 communication).